Finetune LLM(LLAMA) with Huggingface Eco System

LLM Training Pipeline

1. Unsupervised Pretraining ← Self Supervised Learning
2. SFT → Supervised Finetuning
   - → on Parameter Level
     1. full finetuning → train all the Params (weight & Bias)
        - → 1. huge GPU memory 2. multigpu setup
     2. Partial finetuning ← Subset of All Parameter
        - → two method
          1. Old school method — retrain
             - → Freeze all layer and train last O/P layer
             - → Freeze some starting layer & retrain some last layer

LLM ← transformer

CNN based model

early stage LLM

1. BERT
2. T5
3. BART

2. PEFT ← Parameter efficient finetuning — this might work with single GPU and small VRAM
   1. LoRA ← QLoRA ↑ Quantization
      1. Lower Precision model
      2. memory efficient loading
   2. DoRA
   3. Adapter Layer
   4. BitFit
   5. IA^3
   6. Prefix tuning
   7. Prompt tuning

- on data level → Based on the Data Preparation
  1. Non-instructional fine tuning
  2. instructional finetuning

3 Alignment with the human Feedback

Prefrence based learning

alignment of the $\boxed{\text{Responses}}$ ← LLM

1 RLHF ← PPO ← RL

↑ Proximal Prefrence - optimization

OpenAI ←

2 DPO ← Direct Prefrence optimization ← Supervised learning

Question   Response   Feedback
+ve | -ve

Pretraining (Unsupervised or Self Supervised Learning)
Sources of data:
documentation
research papers(ArXiv / PubMed (scientific & academic papers))
Common Crawl data(large-scale web data)
Wikipedia (English + other languages)
Books (Book Corpus)

Objective: Next-token prediction (language modeling).

This process is unsupervised or self-supervised.

Result: A base model is obtained — for example, LLaMA-2-7B, 13B, 70B , Mistral, GPT, Deep seek, Gemini.

After this stage, the model develops knowledge and grammar understanding,

but it still lacks the ability to follow instructions, maintain a polite tone, or produce structured answers effectively.

Normal Fine-Tuning (Domain Adaptation)

Dataset: Plain text (company documents).

Use case: Making LLM a domain expert for areas such as medical, legal, or financial documents.

Goal: To improve the model's understanding and knowledge of a specific domain or language style.

Output style: Produces text continuation — not necessarily capable of following instructions.

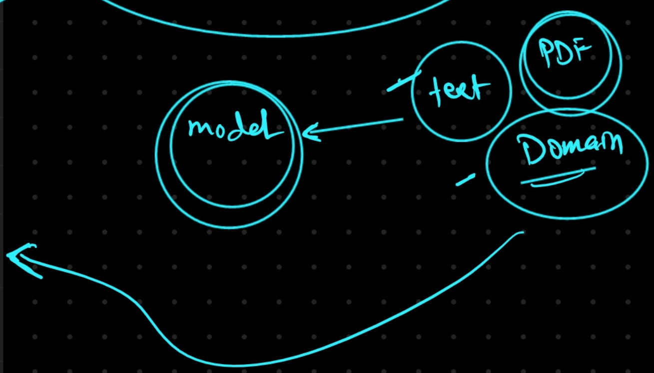
Meaning: The model is learning the statistical distribution of text — not question-answering behavior.

Example (normal fine-tuning dataset row):
Sunny Savita here AI meets unstoppable passion Turning code, creativity, and courage into a legacy. Teaching thousands to master AI, one video at a time. Not just a creator a mentor, a visionary, a job-maker. Don't chase opportunities build them.

PDF

Input (model ko diya gaya text)	Output (model se expected next text)
"Sunny"	"Savita"
"Sunny Savita"	"—"
"Sunny Savita —"	"where"
"Sunny Savita — where"	"AI"
"Sunny Savita — where AI"	"meets"
"Sunny Savita — where AI meets"	"unstoppable"
"Sunny Savita — where AI meets unstoppable"	"passion"
"Sunny Savita — where AI meets unstoppable passion"	"Turning"
"Sunny Savita — where AI meets unstoppable passion. Turning"	"code"
"Sunny Savita — where AI meets unstoppable passion. Turning code"	"creativity"

≠ Non instructional finetuning



Instruction Fine-Tuning (IFT)

Dataset: Follows Instruction + Response format.

Use case: Chatbots, Q&A systems, tutors, coding assistants, customer support, etc.

Goal: Teach the model to follow human instructions — turning it into a chatbot or AI assistant.

Output style: Direct, helpful, and structured answers — often with reasoning or explanations.

Example (IFT dataset row):
Instruction: "Explain what mitochondria does in one sentence."
Response: "Mitochondria generate energy for the cell through respiration."

Meaning: The model learns to produce direct "Question → Answer" or "Prompt → Completion" outputs.
Common datasets: Alpaca, ShareGPT, Dolly, OpenOrca.

Instruction Fine-Tuning (SFT) – Public Conversational Data
Reddit (subreddits like r/AskReddit, r/explainlikeimfive, r/askscience)
Quora (human Q&A style)
StackOverflow / StackExchange (technical question–answer patterns)
GitHub Issues / Discussions (coding and troubleshooting threads)
Open forums & community boards (Hugging Face, Kaggle, etc.)

Combo Strategy (Common in Industry)
Step 1: Normal Fine-Tuning → Adapt LLaMA to your domain (e.g., pharmaceutical documents).
Step 2: Instruction Fine-Tuning → Train it on user question-answer or task instruction data.
(Optional) Step 3: RLHF → Refine using real user feedback and preference ranking.

In Summary:
If the goal is to add knowledge, use normal fine-tuning.
If the goal is to create a chatbot or assistant, use instruction fine-tuning.
If both are desired → perform domain fine-tuning first, then instruction fine-tuning.

Alignment with Human Feedback (RLHF / DPO / RLAIF)
Data: Pairs of responses ranked by humans (or AI models simulating human preference).
Algorithms:
RLHF — Reinforcement Learning from Human Feedback (PPO-based).
DPO — Direct Preference Optimization.
RLAIF — Reinforcement Learning from AI Feedback.
Goal: To make the model polite, safe, helpful, and aligned with human values.
Examples:
GPT-4 = Pretrained GPT + Supervised Fine-Tuning (SFT) + RLHF.
Gemini (Google) = SFT + RLHF + Multimodal Alignment.
DeepSeek R1 = Used reinforcement-style fine-tuning with preference data.

RLHF / DPO Data Sources (Preference Alignment Stage)
Real ChatGPT user logs →
Collected with user consent & anonymization.
Used to see which style of answer users like more.
Human labeler feedback →
Worldwide contractors (Kenya, Philippines, US, etc.) rank "better vs worse" responses.
Helps train the reward model or preference pairs.

Family-Wise Breakdown

1. LLaMA (Meta)
   - Stage 1: Pretraining on trillions of tokens (CommonCrawl, books, code).
   - Stage 2: SFT → made LLaMA-2-Chat.
   - Stage 3: RLHF → polished chat alignment.
   - Public: Meta released both base + chat variants.

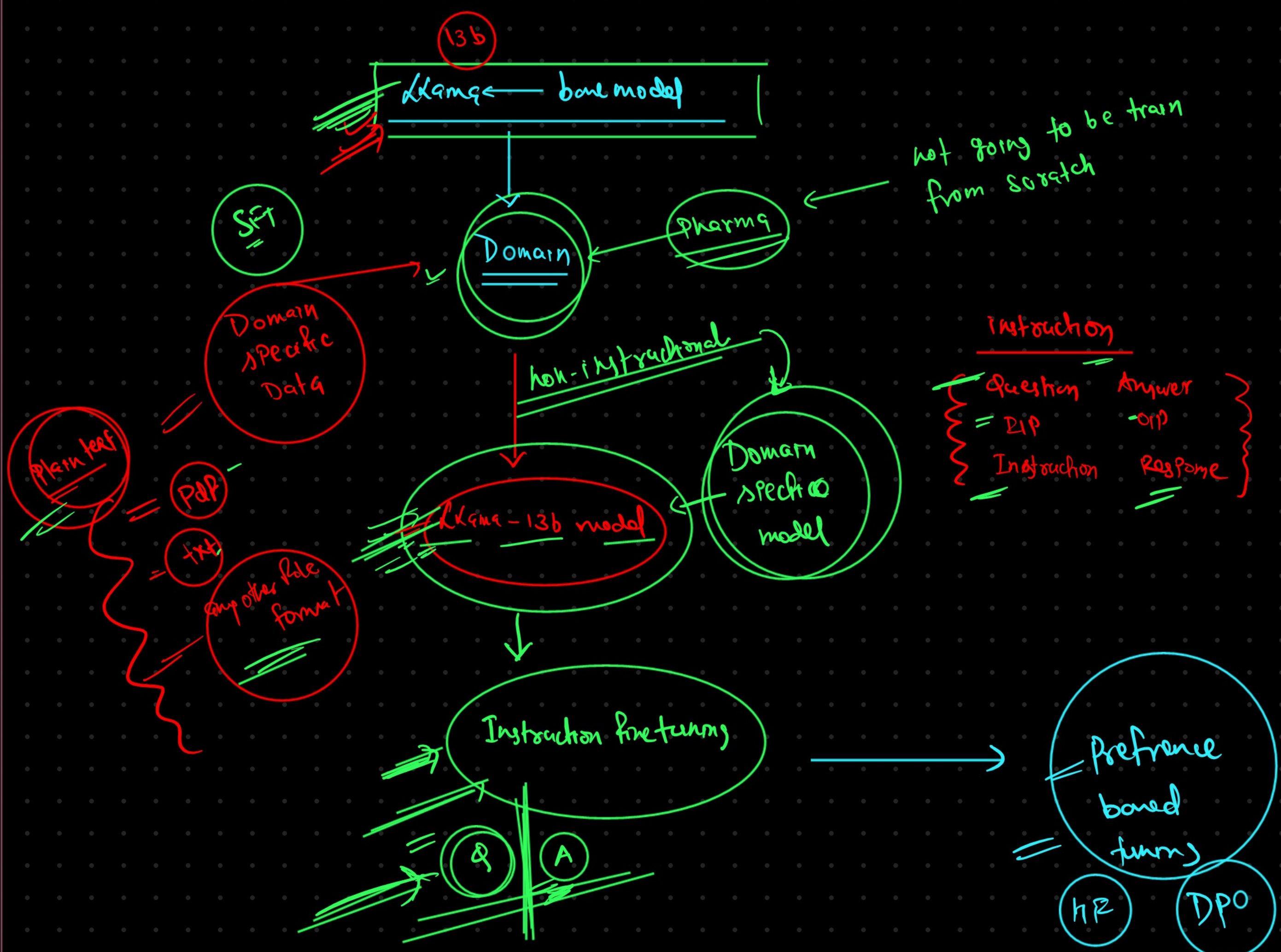
So you can choose base (raw) or chat (aligned).

2. GPT (OpenAI)
   - GPT-3 (2020) → only pretrained (base).
   - InstructGPT (2021) → SFT + RLHF.
   - GPT-3.5 / GPT-4 (2022–23) → heavily aligned with RLHF + safety filters.
   - GPT models are closed-source, so you don't get base weights.

3. Mistral (Mistral AI)
   - Mistral-7B (2023) → pretrained base model.
   - Mixtral-8x7B (MoE, 2023) → pretrained MoE model.
   - Instruct versions (e.g., Mistral-7B-Instruct) → extra SFT done.
   - Some models also aligned with RLHF / DPO.
   - Open weights released (very popular in open-source).

4. DeepSeek (China)
   - Focus: efficient training + domain specialization (coding, reasoning).
   - Pretrained large models (7B, 32B, 67B).
   - Released DeepSeek-Coder → pretrained + SFT on code data.
   - Released DeepSeek-R1 → trained with reinforcement signals (reasoning traces, self-improvement).
   - Open weights released (China's big open-source push).

5. Gemini (Google DeepMind)
   - Pretraining: massive text + code + multimodal data (images, audio, video).
   - SFT: instruction datasets, code, reasoning.
   - RLHF: heavy alignment for safety + multimodal reasoning.
   - Multimodal first-class citizen (not bolted on like GPT-4).
   - Closed-source (like GPT).



1 UPT

2 SFT → NIFT

→ IFT

3 RLHF

DPO

gpt

↓

chatgpt

our own company